

Ricardo Moura Vargas | Senior Software Engineer

Website: <https://ricardomvargas.github.io/>

GitHub: <https://github.com/ricardomvargas>

Linkedin: <https://www.linkedin.com/in/ricardo-m-vargas/>

Phone: + 31 6 4176 2320

E-mail: mvargas.ricardo@gmail.com

SUMMARY

I'm a Senior Software Engineer specializing in web development and frontend architecture, with strong expertise in JavaScript, TypeScript and React. Throughout my career, I've worked across backend, full stack and frontend roles, helping teams modernize legacy systems, define technical solutions, improve developer experience and deliver scalable, user-focused products.

While frontend remains my strongest expertise, I enjoy working across the software stack and continuously expanding my backend knowledge through personal projects in Python and volunteer work developing WordPress and PHP-based websites.

I enjoy collaborating with Product and UX teams, investigating complex technical challenges and contributing to architecture decisions that improve maintainability and long-term scalability. I also leverage AI-assisted software engineering workflows to accelerate development, technical investigations, software design, documentation, implementation and developer productivity.

CORE STRENGTHS

- Frontend & Software Architecture
- JavaScript, TypeScript & React
- Legacy Modernization
- Scalable Web Platforms
- Design Systems & Component Libraries
- Technical Discovery & Solution Design
- Product & UX Collaboration
- Developer Mentoring
- AI-Assisted Software Engineering

TECHNICAL SKILLS

Frontend

React, JavaScript, TypeScript, HTML, CSS

Backend

Python, PHP, .NET/C#

Database

SQL, Microsoft SQL Server, PostgreSQL, SQLite

State Management & Data Fetching

TanStack Query, Redux, Redux Toolkit, REST APIs, Axios

Frontend Architecture & UI

Web Components, Lit, Storybook, Styled Components, Material UI, Bootstrap, Sass

AI-Assisted Software Engineering

ChatGPT, Claude, Claude Code, LLM-based coding assistants, Context Engineering, Custom AI Skills, AI Agents

Mapping & Visualization

OpenLayers, Three.js

Testing

Vitest, Jest, React Testing Library

Build Tools & Tooling

Wordpress, Vite, Webpack, ESLint, npm, pnpm, Yarn, Git

Collaboration & Delivery

Scrum, Kanban, Jira, Figma, Bitbucket

LANGUAGES

Portuguese — Native

English — Professional Proficiency

Dutch — Currently Learning

WORK EXPERIENCE

Senior Software Engineer | Feb. 2023 - present | Cyclomedia B.V.

Working across both Cyclomedia's legacy and next-generation geospatial platforms, contributing to feature development, platform modernization and frontend architecture initiatives. Collaborated closely with Product, UX and engineering teams to investigate complex legacy functionality, define technical solutions and implementation strategies, modernize the platform and improve developer experience through scalable and maintainable frontend solutions.

Key Contributions:

- Helped build a large-scale TypeScript monorepo consisting of multiple packages and shared libraries, using technologies such as React, Vite, Web Components, Lit, Three.js, OpenLayers, TanStack Query, TanStack Table, Vitest, Storybook, ESLint and Amplitude.
- Participated in the migration of a complex legacy application to a modern architecture, analyzing existing functionality and redesigning solutions to improve maintainability, scalability and developer experience.
- Developed geospatial features by integrating WFS, WMS, GeoJSON and ESRI data sources into OpenLayers-based applications.
- Implemented and maintained OpenLayers-based mapping features, managing multiple geospatial layer types and data sources to deliver interactive map visualization capabilities.
- Implemented spatial data transformations and coordinate system reprojection using different Spatial Reference Systems (SRS), ensuring accurate visualization and interaction across multiple geospatial datasets.
- Conducted a deep analysis of the platform's measurement tools, one of its most critical and technically complex features, producing detailed technical documentation, defining implementation strategies and contributing to the delivery of the new solution.
- Led investigations of legacy features, documenting business rules, technical behaviors and implementation details to support the migration process.
- Investigated and proposed the architecture of a framework-agnostic component package based on Web Components, aligned with the company's Design System and intended for use across multiple applications.
- Actively participated in feature discovery, brainstorming sessions and UX discussions, influencing product and interface decisions while balancing business requirements, usability and technical feasibility.
- Worked with Three.js, imagery and LiDAR data to develop and maintain interactive panoramic visualization features for large-scale geospatial applications.
- Introduced AI-assisted software engineering practices by creating reusable project contexts, custom AI skills and specialized agents that streamlined software development, accelerated technical research, improved knowledge sharing and increased developer productivity across day-to-day engineering activities.

Senior Frontend Developer | Feb. 2021 to Feb. 2023 | Paraná Banco

Worked on digital lending solutions for public sector employees, contributing to both new product development and the modernization of existing applications. Collaborated closely with UX designers and frontend teams to improve user experience, evolve the company's design system and deliver scalable React-based solutions.

Key Contributions:

- Contributed to the development of a new lending platform from the ground up, helping define and implement user-facing features and business workflows.
- Developed and maintained financial applications used in the loan origination and loan portability processes.
- Modernized legacy applications by migrating interfaces to the company's design system and improving consistency across products.
- Contributed to the evolution and maintenance of the frontend component library, helping ensure reusability and consistency across multiple applications.
- Collaborated with UX designers to define new features, refine user journeys and translate business requirements into intuitive user interfaces.
- Participated in code reviews, testing activities and frontend best-practice discussions through the company's Frontend Chapter.
- Worked closely with cross-functional teams during planning, refinement and delivery activities within an Agile/Scrum environment.
- Supported deployment and release activities in homologation and production environments.

Frontend Developer | Jan. 2019 to Feb. 2021 | Agrotis

Worked on Agrotis' grain storage management platform, contributing to new feature development, legacy system modernization and frontend architecture improvements. Helped evolve the application by adopting modern React practices, improving maintainability and supporting the company's design system initiatives.

Key Contributions:

- Developed and maintained features for the grain storage module, including the implementation of the services pricing functionality.
- Modernized legacy React applications by migrating class-based components to functional components with Hooks and adopting modern JavaScript (ES6+) practices.
- Contributed to the introduction of TypeScript, helping improve code quality, maintainability and developer experience.
- Collaborated on the adoption and evolution of the company's design system, building reusable UI components and maintaining the shared component library.
- Managed application state using Redux while also simplifying state management approaches when Redux was not required.
- Participated in code reviews and frontend quality initiatives, helping maintain consistency and improve development standards across the application.

Full Stack Developer | Sep. 2014 – Sep. 2015 | PrimeLan Tecnologia

At Primelan I was responsible for creating websites by demand following the web standards and web apps that integrate with WIFI equipment for user authentication. I also created Vitória Online, the official android application for Vitória (ES) city.

Full Stack Developer | Nov. 2012 – Sep. 2014 | Megawork

I helped to build a suppliers pricing application which integrates with SAP. Later I worked building web sites, applications and reports that read data from SAP.

.Net Developer | Feb. 2012 – Nov. 2012 | Phidelis Tecnologia

As a .Net developer, I worked on a web application for managing schools and universities, fixing bugs, creating custom features, doing code reviews and training users to use the platform.

.Net Developer | Jul. 2011 – Feb. 2012 | Superius Consultoria LTDA

At Superius, I was part of a team that was building ERP application using .Net 3.0, C#, DevExpress, Entity Framework and SqlServer.

Junior .Net Developer | Jan. 2010 – Jul. 2011 | Vixtime

At Vixtime I helped to build a News application and News website for the ESHoje News channel, I worked on creating CMS systems and websites by demand following the W3C standards.